

Rubab Ali Qureshi

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EDUCATION

Bachelor of Science in Cyber Security
Ghulam Ishaq Khan Institute (GIKI)

September 2023 – June 2027

PROFESSIONAL EXPERIENCE

SOC Intern
ITSOLERA

July 2025 – September 2025

- Monitored and triaged **1,000+ SIEM alerts weekly** at a cybersecurity operations center, covering endpoint, network, and application threat vectors.
- Investigated suspicious activity by correlating **Wazuh** logs across endpoints, firewalls, and network devices using rule-based and behavioral analysis techniques.
- Refined alert prioritization rules, cutting **false positive rate by ~20%** and reducing mean time to triage per incident from ~15 to ~12 minutes.

Cybersecurity Intern
Nexus Technologies

June 2024 – August 2024

- Conducted vulnerability assessments across **5+ client web applications and internal networks**, covering OWASP Top 10 attack surfaces and network misconfigurations.
- Performed penetration testing using **Nmap, Metasploit, and Burp Suite**; identified and documented **SQL injection, XSS, and privilege escalation** vectors.
- Delivered remediation reports that enabled clients to patch **12+ critical vulnerabilities**, directly reducing exploitable attack surface across production environments.

VOLUNTEER WORK

Hostel Network Head
NETRONIX GIKI

September 2025 – Present

- Managed hostel WiFi network serving **150+ users**; resolved connectivity issues and improved uptime.
- Led a technician team; handled complaints end-to-end and ensured timely resolution.
- Collaborated with IT Department on infrastructure planning and network upgrades.

Network Support Technician
NETRONIX GIKI

October 2023 – September 2025

- Maintained LAN infrastructure; handled crimping, uplink management, flooding issues, and hostel network troubleshooting.

PROJECTS

— **Secure Healthcare System** — Secure Software Design

- Built a security-hardened healthcare records platform for a multi-role clinical environment requiring strict access control and audit compliance.
- Implemented **RBAC, JWT authentication, and MFA**; integrated **Wazuh SIEM, Supabase RLS, and WAF** with rate-limited, secure APIs.
- Eliminated unauthorized access paths across all tested roles, achieving **zero critical vulnerabilities** in post-build penetration testing covering SQLi, XSS, and DoS vectors.

— **AI-Based Phishing Detection** — Flask, React, Machine Learning

- Developed an AI-powered phishing and malicious URL detection system for real-time threat identification in browser and email contexts.
- Trained a classification model on labeled URL datasets using **Scikit-learn**; deployed via a **Flask** backend with a **React** frontend for live URL scanning.
- Achieved **~92% detection accuracy** on test set, correctly flagging phishing URLs that bypassed standard blacklist-based filters.

— **Data Loss Prevention System** — Machine Learning, Rule-Based Systems

- Designed a Data Loss Prevention system to detect and block exfiltration of sensitive data across file transfers and network channels.
- Implemented a hybrid engine combining **ML classifiers** for content sensitivity detection with **rule-based policies** for pattern matching on PII, credentials, and confidential keywords.
- Reduced simulated data leakage incidents by **~85%** in testing, with under **5% false positive rate** on benign file transfers.

— **MachineMind – OCR + LLM Pipeline** — OCR, LLMs

- Built an automated industrial machine identification system to eliminate manual lookup and cataloging of machines from physical labels and nameplates.
- Combined **OCR pipelines** for text extraction from machine images with **LLM-based parsing** to structure raw text into searchable machine records.
- Reduced machine identification time from **~10 minutes manual** to **under 30 seconds**, cutting cataloging effort by over **95%**.

— **Secure LoRa IoT System** — AES Encryption, IoT, Anomaly Detection

- Designed a secure communication framework for an IoT sensor network transmitting sensitive environmental data over long-range **LoRa** channels.
- Implemented end-to-end **AES-128 encryption** for all node-to-gateway packets and built an anomaly detection layer to flag irregular transmission patterns indicating tampering or node compromise.
- Prevented all simulated man-in-the-middle and replay attacks in testing; anomaly detection identified **100% of injected malicious packets** in controlled trials.

— **Honeypots & Threat Analysis** — Wazuh, Elastic Stack

- Set up a honeypot infrastructure to capture and study real-world attacker behavior targeting simulated exposed services (SSH, HTTP, FTP).
- Deployed low-interaction honeypots and ingested attack logs into **Wazuh SIEM** and **Elastic Stack** for correlation, visualization, and TTPs mapping against the MITRE ATT&CK framework.
- Captured and analyzed **500+ attack events** within the first week, identifying brute-force, credential stuffing, and port scanning as the dominant attack patterns.

— **Enterprise Secure Network Design** — VLANs, OSPF, NAT, ACLs, VPN

- Designed a full enterprise network architecture for a multi-department organization requiring segmented access, dynamic routing, and secure remote connectivity.
- Implemented **VLAN segmentation, OSPF dynamic routing, NAT, ACLs, and site-to-site VPN** using **Cisco Packet Tracer**; enforced inter-VLAN access policies at the distribution layer.
- Achieved full department isolation with **zero unintended cross-VLAN access** in simulation; VPN tunnel maintained **100% uptime** across all tested failover scenarios.

SKILLS & CERTIFICATIONS

- **Languages:** Python, C++, JavaScript, SQL, Solidity.
- **Security:** SIEM (Wazuh), SOC Operations, Threat Analysis, Vulnerability Scanning, RBAC, DLP, IDS/IPS, Firewalls, MITM, SQL Injection, DoS/DDoS, Malware Analysis, Cryptography.
- **Networking:** TCP/IP, DNS, DHCP, LAN/WAN, VLANs, OSPF, NAT, VPN, Wireless Networks.
- **Tools:** Wireshark, Nmap, Metasploit, Docker, AWS CLI, Kali Linux.
- **Systems:** Linux, Windows Server, macOS, Containerization, Cloud Security.
- **Development:** React, Node.js, Flask, FastAPI, PostgreSQL, Git.
- **Core:** Penetration Testing, Traffic Analysis, Endpoint Security, Threat & Vulnerability Management.
- **Certifications:** CEH v12 (LearnKartS), Advanced Cybersecurity, IBM Network Security & OS Security.